pandas

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1 PANDAS

Pandas is a very useful tool for 'wrangling' with data. You can learn all about it at https://pandas.pydata.org In this exercise we will just run through some of the basics.

1.1 5.1

Execute the commands below and explain what the code does

[1]:
```
import pandas as pd
```

[2]:
```
url = 'https://raw.githubusercontent.com/justmarkham/DAT8/master/data/chipotle.
↪tsv'
chipo = pd.read_csv(url, sep = '\t')
```

1.2 5.2

Execute the commands below and explain what the above code does. You'll have to execute them one by one.

[3]:
```
chipo.head(10)
```

[3]:

	order_id	quantity	item_name	choice_description	item_price
0	1	1	Chips and Fresh Tomato Salsa	NaN	$2.39
1	1	1	Izze	[Clementine]	$3.39
2	1	1	Nantucket Nectar	[Apple]	$3.39
3	1	1	Chips and Tomatillo-Green Chili Salsa	NaN	$2.39
4	2	2	Chicken Bowl	[Tomatillo-Red Chili Salsa (Hot), [Black Beans, ...	$16.98
5	3	1	Chicken Bowl	[Fresh Tomato Salsa (Mild), [Rice, Cheese, Sour ...	$10.98
6	3	1	Side of Chips	NaN	$1.69
7	4	1	Steak Burrito	[Tomatillo Red Chili Salsa, [Fajita Vegetables, ...	$11.75
8	4	1	Steak Soft Tacos	[Tomatillo Green Chili Salsa, [Pinto Beans, Chee...	$9.25
9	5	1	Steak Burrito	[Fresh Tomato Salsa, [Rice, Black Beans, Pinto B...	$9.25

```
4  [Tomatillo-Red Chili Salsa (Hot), [Black Beans…    $16.98
5  [Fresh Tomato Salsa (Mild), [Rice, Cheese, Sou…    $10.98
6                                                NaN     $1.69
7  [Tomatillo Red Chili Salsa, [Fajita Vegetables…    $11.75
8  [Tomatillo Green Chili Salsa, [Pinto Beans, Ch…     $9.25
9  [Fresh Tomato Salsa, [Rice, Black Beans, Pinto…     $9.25
```

It shows the 10 first observations of the dataset

[4]:
```
chipo.tail(10)
```

[4]:
```
      order_id  quantity            item_name  \
4612      1831         1        Carnitas Bowl
4613      1831         1                Chips
4614      1831         1        Bottled Water
4615      1832         1   Chicken Soft Tacos
4616      1832         1  Chips and Guacamole
4617      1833         1        Steak Burrito
4618      1833         1        Steak Burrito
4619      1834         1   Chicken Salad Bowl
4620      1834         1   Chicken Salad Bowl
4621      1834         1   Chicken Salad Bowl

                                choice_description item_price
4612  [Fresh Tomato Salsa, [Fajita Vegetables, Rice,…     $9.25
4613                                              NaN     $2.15
4614                                              NaN     $1.50
4615   [Fresh Tomato Salsa, [Rice, Cheese, Sour Cream]]     $8.75
4616                                              NaN     $4.45
4617  [Fresh Tomato Salsa, [Rice, Black Beans, Sour …    $11.75
4618  [Fresh Tomato Salsa, [Rice, Sour Cream, Cheese…    $11.75
4619  [Fresh Tomato Salsa, [Fajita Vegetables, Pinto…    $11.25
4620  [Fresh Tomato Salsa, [Fajita Vegetables, Lettu…     $8.75
4621  [Fresh Tomato Salsa, [Fajita Vegetables, Pinto…     $8.75
```

It shows the 10 last observations of the dataset

[5]:
```
chipo.shape[0]
```

[5]:
```
4622
```

It shows the number of rows of the dataset

[6]:
```
chipo.shape[1]
```

[6]:
```
5
```

It shows the number of columns of the dataset

1.3 5.3

Execute the commands below and explain the results. Again, you'll have to do them one by one.

[7]:
```
chipo.columns
```

[7]:
```
Index(['order_id', 'quantity', 'item_name', 'choice_description',
       'item_price'],
      dtype='object')
```

It display the list of the column names

[8]:
```
chipo.dtypes
```

[8]:
```
order_id               int64
quantity               int64
item_name             object
choice_description    object
item_price            object
dtype: object
```

It displays the data type of each column

1.4 5.4

Explain what the command below does

[11]:
```
chipo.item_name.nunique()
```

[11]:
```
50
```

It display the number of unique values in the item_name column

1.5 5.5

Explain what the command below does.

[12]:
```
chipo.describe()
```

[12]:

	order_id	quantity
count	4622.000000	4622.000000
mean	927.254868	1.075725
std	528.890796	0.410186
min	1.000000	1.000000
25%	477.250000	1.000000
50%	926.000000	1.000000
75%	1393.000000	1.000000
max	1834.000000	15.000000

It gives a brief description of the dataset with the basic statistics (mean, median, std, count, min, max) of each column

1.6 5.6

Finally, explain what the commands below do.

[13]:
```
newchip=chipo.fillna(-1)
```

It impute the NA of the dataset with the value -1

[15]:
```
newchip.head(10)
```

[15]:

	order_id	quantity	item_name	choice_description	item_price
0	1	1	Chips and Fresh Tomato Salsa	-1	$2.39
1	1	1	Izze	[Clementine]	$3.39
2	1	1	Nantucket Nectar	[Apple]	$3.39
3	1	1	Chips and Tomatillo-Green Chili Salsa	-1	$2.39
4	2	2	Chicken Bowl	[Tomatillo-Red Chili Salsa (Hot), [Black Beans…	$16.98
5	3	1	Chicken Bowl	[Fresh Tomato Salsa (Mild), [Rice, Cheese, Sou…	$10.98
6	3	1	Side of Chips	-1	$1.69
7	4	1	Steak Burrito	[Tomatillo Red Chili Salsa, [Fajita Vegetables…	$11.75
8	4	1	Steak Soft Tacos	[Tomatillo Green Chili Salsa, [Pinto Beans, Ch…	$9.25
9	5	1	Steak Burrito	[Fresh Tomato Salsa, [Rice, Black Beans, Pinto…	$9.25

It displays the 10 first rows of the new dataset